MIL-D-51048F(EA)
6 August 1976
SUPERSEDING
MIL-D-51048E(MU)
13 July 1972

MILITARY SPECIFICATION

DECONTAMINATING APPARATUS,

PORTABLE, DS2, 1-1/2 QUART, M11

This specification is approved for use by Edgewood Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of decontaminating apparatus.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, Flat and Seals.
PPP-B-26 - Bag, Plastic (General Purpose).
PPP-B-601 - Box, Wood, Cleated-Plywood.
PPP-B-621 - Box, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping, Fiberboard.

MILITARY

MIL-B-46506 - Box, Ammunition Packing, Wood, Wirebound.

STANDARDS

FEDERAL

Fed. Std. No. 601 - Rubber: Sampling and Testing.
Fed. Std. No. 356 - Commercial Packaging of Supplies and Testing.

FSC 4230

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

DL5-51-269 - Decontaminating Apparatus, Portable, DS2, 1-1/2 Quart, ABC-M11.
DL5220-A50-9759 - Fixed Dial Indicator.
D5220-A50-9756 - Fixed, Location.
C5220-A50-9757 - Plug, Thread Setting.
DL5220-A50-9758 - Fixture, Dial Indicator.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D5-51-269 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the decontaminating apparatus shall conform to the specifications and drawings listed on Drawing D5-51-269 and subsidiary drawings.

3.2 Assembly. The decontaminating apparatus shall be assembled as shown on Drawing D5-51-269.

3.3 Container. The container, Drawing D5-51-302, when subjected to a hydrostatic pressure of 375-15 pounds per square inch gage (psig) for a period of 60 ± 5 seconds, shall not leak when tested as specified in 4.4.4.1.

3.4 Spray head assembly. The spray head assembly (less container and bracket) when subjected to an air pressure of 240 ± 5 psig, shall

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not show a pressure drop greater than 2-1/2 psig in a period of 60 ± 5 seconds when tested as specified in 4.4.4.2.

3.5 Burst. The empty container (see 3.3) shall not leak or burst when subjected to an internal hydrostatic pressure of 720 psig minimum for a minimum period of 30 seconds when tested as specified in 4.4.4.3.

3.6 Functioning.

3.6.1 Empty. The decontaminating apparatus shall puncture the nitrogen-filled cylinder and discharge the nitrogen through the nozzle. All operational parts shall function freely when tested as specified in 4.4.4.4(a).

3.6.2 Filled. The decontaminating apparatus, filled with DS2 (see 6.2) or propylene glycol as applicable (see 4.3.3.1(b) and 4.4.4.4(b), shall produce a spray which is full and symmetrical. When fully pressurized, the spray shall carry a minimum distance of six feet under ambient temperature and zero wind speed. The spray pattern at the above distance shall be a minimum of 10 inches in diameter when tested as specified in 4.4.4.4(b).

3.7 Volumetric change in O-rings. Volumetric change as specified on Drawing B5-51-371 shall not exceed 12 percent when tested as specified in 4.4.4.5.

3.8 Body soundness. The soundness of the body shall conform to Drawing D5-51-291 when tested as specified in 4.4.4.6.

3.9 Preproduction. Prior to the start of regular production, a pre-production lot of decontaminating apparatuses shall be produced in accordance with this specification for examination and tests (see 4.3).

3.10 Workmanship. The decontaminating apparatus shall be free from foreign matter (oil, grease or dirt) and damage such as dents, cracks, burrs, sharp edges or deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contact or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Govern-

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ment reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3)
(b) Quality conformance isnpection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot.

4.3.1.1 Decontaminating apparatus. A preproduction lot of 100 sets of components shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production. Components shall be assembled to meet the lot quantities specified in table I.

4.3.1.2 O-Rings. Including the above, para 4.3.1.1, a preproduction lot of 200 of each type O-Ring conforming to Dwg B5-51-371 shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination and nondestructive tests. Sampling shall be conducted at random in accordance with table I in the sequence listed.

4.3.2.2 For destructive tests.

(a) Burst. A sample of 38 containers shall be taken at random from the preproduction lot. Twenty-two for test and 16 for retest.

(b) O-rings. A sample of 30 o-rings shall be taken at random from the preproduction lot.

4.3.3 Inspection procedure.

4.3.3.1 For examination and nondestructive tests.

(a) Components. Each component shall be examined and tested for all the drawing requirements, the acceptance number is zero.

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TABLE I - Sampling

Component/assembly	Quantity: Lot	Quantity: Sample	Inspection procedure
Each component	100	5	4.3.3.1(a)
Container	100	22/38	4.3.3.2(a)
O-Rings	200 ea	30 ea	4.3.3.2(b)
Body	100	2	4.3.3.2(c)
Spray head assembly	98	30	4.3.3.1(b)
Container	62/78*	62/78*	4.3.3.1(b)
Nozzle	100	30	4.3.3.1(b)
Apparatus	62/78*	30	4.3.3.1(b)
			4.3.3.1(c)
Preparation for delivery	62/78*	30	4.3.3.1(b)

*Of the original 100 preproduction lot of containers those remaininig after destructive testing, 4.3.3.2(a), shall be used to meet the remaining requirements.

(b) Decontaminating apparatus, container, spray head assembly, nozzle and preparation for delivery. Each sample item shall be examined and tested in accordance with the classification of defects (4.4.3.4), except major defect 105 of 4.4.3.4(a) is not applicable. The AQL's do not apply. The acceptance number is zero.

(c) Decontaminating apparatus (filled). The sample apparatus, examined and tested in accordance with 4.3.3.1(b), if accepted, shall be tested in accordance with 4.4.4.4(b) except that the container shall be filled with DS2 from the 1-1/3 quart can (see 6.2 and 6.3). The acceptance number is zero.

4.3.3.2 For destructive tests.

(a) Burst. The 22 sample containers shall be tested in accordance with 4.4.4.3. If one container fails, retest using the remaining 16 containers. If two containers fail in test and retest the lot shall be rejected.

(b) O-rings. The sample O-rings shall be tested in accordance with 4.4.4.5. The acceptance number is zero.

(c) Body. The sample bodies shall be tested in accordance with 4.4.4.6. The acceptance number is zero.

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4.3.4 Acceptance/rejection criteria. The preproduction lot shall comply with the examinations and tests specified in 4.3.3 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting.

4.4.1.1 O-rings. A lot shall consist of the O-rings produced by one manufacturer, from the same materials, under essentially the same manufacturing conditions. However, no more than one batch of material (see 6.5) shall be represented in any one lot of finished O-rings.

4.4.1.2 Decontaminating apparatus. A lot shall consist of the apparatus produced by one manufacturer, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of O-rings shall be represented in any one lot of finished apparatus.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests.

(a) Decontaminating apparatus (empty). Sampling shall be conducted in accordance with MIL-STD-105.

(b) Decontaminating apparatus (filled). Sampling shall be conducted in accordance with MIL-STD-105, level S-3.

4.4.2.2 For destructive tests.

(a) Burst. One container shall be taken at random from every 500 containers.

(b) O-ring. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

(c) Body soundness. Sampling shall be in accordance with MIL-STD-105, level S-3.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests.

(a) Decontaminating apparatus (empty). The sample apparatus and the level A preparation for delivery shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

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(b) Decontaminating apparatus (filled). The sample apparatus shall be tested in accordance with 4.4.4.4(b) and MIL-STD-105, using an AQL of 1.5 percent defective.

4.4.3.2 For destructive tests.

(a) Burst. The sample container shall be tested in accordance with 4.4.4.3. The acceptance number is zero.

(b) O-ring. The sample O-rings shall be tested in accordance with 4.4.4.5 and MIL-STD-105, using an AQL of 1.0 percent defective.

(c) Body soundness. The sample bodies shall be tested in accordance with 4.4.4.6 and MIL-STD-105, using an AQL of 1.0 percent defective.

4.4.3.3 For critical defects. Each item in the lot shall be tested for the critical characteristic listed in the classification of defects.

4.4.3.4 Classification of defects.

(a) Decontaminating apparatus, portable, DS2, 1-1/2 quart, ABC-M11 (Drawing D5-51-269).

Categories		Defects	Acceptance standards
Critical:		None defined	
Major:	AQL	1.0 percent defective	
101		Component missing	
102		Component incorrectly assembled	
103		Vapor phase inhibitor missing	
104		Name plate marking incorrect	
*105		Functioning, empty	4.4.4.4(a)
106		Workmanship (3.9)	
Minor:	AQL	4.0 percent defective	
201		Mounting bracket loose	
202		Protective finish missing	
203		Fill line missing	

*Not applicable for preproduction inspection.

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4.4.3.4 Classification of defects (continued).

(b) Container, assembly and detail (Drawing D5-51-302).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.4.4.1
Major:	AQL 1.0 percent defective	
101	Thread (1-7/8-12UN-2B) incorrect	
102	Parallelism incorrect	
103	Machine finish incorrect (Drawing B5-51-289)	
104	Workmanship (3.9)	

(c) Spray head, assembly (Drawing D5-51-314).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Leakage	4.4.4.2
102	Thread (1-7/8-12UN-2A) incorrect (Drawing D5-51-291)	
103	Perpendicularity incorrect (Drawing D5-51-291)	
104	Machine finish incorrect (Drawing D5-51-291)	
105	Workmanship (3.9)	

(d) Nozzle (Drawing D5-51-310).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Thread incorrect
102	Diameter (0.061) incorrect
103	Diameter (0.249) incorrect

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4.4.3.4 Classification of defects (continued).

(d) Nozzle (Drawing D5-51-310)(continued).

Categories	Defects	Acceptance standards
Major(continued):		
104	Slot width (0.031) incorrect	*DL5220-A50-9759
105	Slot depth (0.030) incorrect	
106	Location slots and diameter (0.249) incorrect	*D5220-A50-9756
107	Thickness (0.012) prior to drilling (0.061) hole incorrect	*C5220-A50-9758
108	Thickness (0.070) prior to forming slots incorrect	*DL5220-A50-9758
109	Workmanship (3.9)	

*Use of this gage design is mandatory.

(e) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Package loose
102	Packing loose
103	Strapping loose
104	Marking missing or illegible

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Container. Subject the container to the specified hydrostatic pressure for a period of 60 ± 5 seconds (see 6.4). The gages shall be accurate to within 1 percent. Thoroughly dry the container after test.

4.4.4.2 Spray head. The spray head assembly, less the nitrogen cylinder, shall be attached to a pressurization fixture of 1-1/2 quart maximum capacity (container, Drawing D5-51-302, may be used) fitted with a pressure gage, valve and means of pressurizing with air or nitrogen. The fixture shall be pressurized to the specified pressure and the valve closed between the pressure source and fixture. Record the pressure drop after the specified time.

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4.4.4.3 Burst. Subject the container to the specified pressure for the required period of time. Increase the pressure until the container leaks or ruptures and record the pressure (see 6.4).

4.4.4.4 Functioning.

(a) Empty. Test the empty decontaminating apparatus in the following manner:

(1) Assemble the spray head to an empty container.
(2) Place a nitrogen-filled cylinder in the spray head.
(3) Pull the arming pin and slowly lift the handle.
(4) Aim and press top in 5 second bursts (5 seconds between burst and check valve closing) until all the nitrogen is expended. Check ease of operation.

(b) Filled. Test the filled decontaminating apparatus in the following manner:

(1) Fill the container with a 1-1/3 quart mixture of propylene glycol (40 percent by volume) and water (60 percent by volume).
(2) Assemble the spray head to the container and place a nitrogen-filled cylinder in the spray head.
(3) Pull the arming pin and slowly lift the handle, until it locks.
(4) Fasten the decontaminating apparatus to a fixed horizontal surface so that center line of nozzle is approximately 4 feet from ground. The nozzle shall be 6 feet ± 6 inches from a vertical target, and the center line of the nozzle shall be perpendicular to the plane of the target. Press down on lever, discharging the liquid for a period of 5 seconds. The spray pattern shall be a minimum of 10 inches in diameter. The above procedure shall be done with zero wind speed.
(5) Repeat the 5-second bursts, pausing 5 seconds between bursts, until all pressure is exhausted.
*(6) Measure and record diameter of spray pattern.
(7) Thoroughly clean and dry the decontaminating apparatus after test.

*If the diameter is less than that specified, repeat 4.4.4.4(b)(6). If the apparatus fails the test the second time, that apparatus shall be one defect.

4.4.4.5 Volumetric change in O-rings. Volumetric change (swelling) shall be determined in accordance with method 6211 of Federal Test Method Standard No. 601, except that the specimen shall be qualification sample O-rings. The period of immersion shall be a minimum of 4 days. The

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fluid shall be DS2 (see 6.2) and maintained at a temperature of 70° ± 2° C (158° ± 3.6° F).

4.4.4.6 Body Soundness. Sample castings shall be sectionalized and examined for soundness.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Level shall be as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Mounting hardware. Three (3) each nuts (MS51967-2), bolts (MS90725-8), and lockwashers (MS35340-44) shall be inclosed in a barrier bag conforming to PPP-B-26, type II, style 1, .004 inches thick material, having inside dimensions of 2 inches by 3 inches. Closure of the bag shall be accomplished by heat sealing in accordance with bag suppliers recommendations for temperature, dwell, and pressure requirements.

5.1.1.2 Decontaminating apparatus. Each decontaminating apparatus with attached holding bracket containing two (2) nitrogen cylinders, and the packaged mounting hardware, in accordance with 5.1.1.1, shall be packaged in a fiberboard box conforming to RSC, W5c or V3c of PPP-B-636, having inside dimensions of 5 inches by 4-1/8 inches by 14 inches, length, width, and depth, respectively. Filler pads of the same type material as the box shall be used in the top and sides, inside the box as necessary to restrict free movement of the item. The box shall be closed in accordance with method IV in the appendix to PPP-B-636.

5.1.2 Level B. The decontaminating apparatus with attached holding bracket and two nitrogen-filled cylinders shall be packaged in accordance with 5.1.1, except the box and filler pads shall be class domestic, grade 275 minimum of PPP-B-636 and taping of top seams and joints is not required. Boxes shall be closed in accordance with method I in the appendix to PPP-B-636.

5.2 Packing. Level shall be as specified (see 6.2).

5.2.1 Level A. The apparatus, packaged in accordance with 5.1.1, shall be packed in containers conforming to the following:

(a) Box, ammunition packing wood, wirebound, type I, class 2, grade A, of MIL-B-46506.

(b) Box, class 2, style 4, grade A, 100-250 pound load limit for type 1 or 2 load of PPP-B-621.

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(c) Box, wood, cleated-plywood, overseas type, style I, 0-100 pound load limit, type 1 and 2 loads, using PS-1 standard plywood of PPP-B-601. The inside dimensions of the boxes shall be 18 inches by 16-1/4 inches by 14-3/4 inches, length, width, and depth, respectively. Arrangement of the packages in the box shall be three rows of four each. Filler pads of weather resistant fiberboard shall be included, as necessary, to restrict free movement of the packages. The boxes shall be closed and strapped in accordance with the appendix of their respective specification. (Strapping is not required on containers conforming to MIL-B-46506.) Strapping and seals shall conform to QQ-S-781 [Strapping - type I (Regular duty), finish B, grade 2, and seals - type D, styles I or IV, finish B, grade 2].

5.2.2 Level B. The apparatus, packaged as specified, shall be packed as specified in 5.2.1, except a style RSC, grade V3c fiberboard box conforming to PPP-B-636 shall be used. The box shall be closed in accordance with method V closure requirements and reinforced using nonmetallic or tape banding in accordance with reinforcing requirements in the appendix to PPP-B-636.

5.2.3 Commercial. Commercial packing shall be as specified in FED-STD-356.

5.3 Palletization. When specified in the contract or order (see 6.2), the apparatus shall be palletized in accordance with Drawing D5-51-328.

5.4 Marking. In conjunction with pertinent information provided in the contract or order, marking shall be as follows:

5.4.1 Packaging. Packages, 5.1.1.1 and 5.1.1.2, shall be marked in accordance with MIL-STD-129.

5.4.2 Shipping (exterior) container, levels A & B. Boxes shall be marked in accordance with Drawing D5-51-378. Address marking is not required on individual boxes when palletized or shipped in full carload or truckload lots from a single consignor to a single consignee.

5.4.3 Shipping (exterior) container commercial. Boxes shall be marked in accordance with FED-STD-356.

5.4.4 Palletization. Palletized loads shall be marked in accordance with Drawing D5-51-328.

6. NOTES

6.1 Intended use. This item is intended for use in the field for spraying DS2 decontaminating agent. It can be hand carried and refilled in the field.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packaging and packing required.
(c) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

(d) When palletization is required.
(e) The DS2 (MIL-D-50030) will be Government furnished.

6.3 Component specification. The component specification for this item is as follows:

MIL-C-51098 - Cylinder, Nitrogen-Filled, Puncturable Seal.

6.4 Caution. Suppliers are hereby notified to provide adequate protection for personnel during testing with high pressures and DS2.

6.5 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.6 International interest. Certain provisions of this specification are subject of international standardization agreement (Quadripartite Standardization Agreement QSTAG No. 235). When amendments, revision or cancellation of this specification are proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4230-A054

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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